V.L Memorial Public School

475 Vijay Nagar

CHAPTER-15-QUESTIONS-CLASS7TH-MATHS

Class 07 - Maths (Ganita Prakash)

Section A

1. If $4x - 10 = 14$, then $x =$ **[1]**

   a) 4 b) 5

   c) 6 d) 7

2. A bag weighs the same as three 2-kg weights and one 1-kg weight. **[1]**
   What is the weight of the bag?

   a) 5 kg b) 8 kg

   c) 6 kg d) 7 kg

3. If 3x-7=11, then x=..... **[1]**

   a) 5 b) 6

   c) 7 d) 4

4. In ancient Indian notation, known quantities were called: **[1]**

   a) rū b) yā

   c) kā d) nī

5. The value of x in 5 x=20 is **[1]**

   a) 4 b) 3

   c) 5 d) 2

6. The solution of 2 x+9=19 is **[1]**

   a) 5 b) 3

   c) 4 d) 6

7. The word “algebra” comes from the Arabic word used in the book of: **[1]**

   a) Al-Khwarizmi b) Aryabhata

   c) Bhaskara d) Brahmagupta

8. Five identical bottles together weigh 20 kg. **[1]**
   Which equation represents this situation correctly?

   a) b + 5 = 20 b) 5b = 20

   c) 5 + b = 20 d) b = 20 - 5

9. Which correction is needed? **[1]**
   5x = 25
   x = 25 - 5

a) No change needed

b) Change 25 to 20

c) Change minus to divide

d) Change minus to multiply

10. Solve: 5y - 10 = 15 **[1]**

a) 3

b) 5

c) 6

d) 4

11. Three identical boxes and one 4-kg weight together weigh 16 kg. **[1]**
What is the weight of one box?

a) 5 kg

b) 2 kg

c) 3 kg

d) 4 kg

12. Solve: 4(m - 2) = 12 **[1]**

a) 5

b) 4

c) 6

d) 3

13. Solve: 2a + 7 = a + 15 **[1]**

a) 7

b) 6

c) 8

d) 9

14. Find the mistake in the following: **[1]**
2x + 5 = 15
2x = 15 + 5
x = 10

a) Adding 5 instead of subtracting

b) Dividing by 2 is wrong

c) There is no mistake

d) Writing 2x is wrong

15. A student solves: **[1]**
6p = 18
p = 18 - 6
p = 12
The correct value of p is:

a) 3

b) 12

c) 2

d) 6

16. On a balance scale, 3 identical boxes balance with 15 identical balls. **[1]**
If one ball weighs 2 units, what is the weight of one box?

a) 10

b) 5

c) 12

d) 8

17. A student writes: **[1]**
3y + 4 = 13
3y = 13 - 4
y = 3
Which statement is correct?

a) Only second step is wrong

b) Only first step is wrong

c) The final answer is wrong

d) No step is wrong

18. Two identical packets weigh the same as 4 kg and 6 kg together. **[1]**
What is the weight of one packet?

a) 6 kg

b) 4 kg

c) 10 kg

d) 5 kg

19. A student solves: **[1]**
4x - 6 = 10
4x = 16
x = 4
Which step is incorrect?

a) 4x = 16

b) x = 4

c) 4x - 6 = 10

d) No step is incorrect

20. Solve: 3x + 5 = 20 **[1]**

a) 6

b) 4

c) 3

d) 5

21. **Assertion (A):** Adding the same number to both sides of an equation keeps the equality true. **[1]**
Reason (R): Addition and subtraction are inverse operations.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

22. **Assertion (A):** Bījagaṇita is the ancient Indian term used for algebra. **[1]**
Reason (R): The word *bīja* means seed, and algebra helps in discovering unknown quantities step by step.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

23. **Assertion (A):** The solution of $8 - 4x = 2(x - 3) - (x - 5)$ is $x = 3$. **[1]**
Reason (R): When we move a variable term from one side of an equation to the other, its sign changes.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

24. **Assertion (A):** If three identical boxes balance with 12 identical balls, then one box is equal to 4 balls. **[1]**
Reason (R): Each box has equal weight and total balls are equally distributed among the boxes.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

25. **Assertion (A):** The solution of the equation $4x = 16$ is $x = 4$. **[1]**

Reason (R): We subtract 4 from both sides to remove the coefficient of x.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

26. **Assertion (A):** In the equation $6(p + 3) = 18$ dividing both sides by 6 gives $p + 3 = 3$. **[1]**

Reason (R): When a variable term is multiplied by a constant, we can divide both sides by that constant to isolate the variable.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

27. **Assertion (A):** In the equation $6x = 18$, writing $x = 18 - 6$ is a correct step. **[1]**

Reason (R): To remove multiplication by 6, we must divide both sides by 6.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

28. Four identical bottles together weigh 24 kg. **[1]**

Find the weight of one bottle.

29. In the equation **[1]**

$7 - x = 3$,

find the value of x.

30. A student solves: **[1]**

$4x = 16$

$x = 16 - 4$

Write the correct value of x.

31. Solve: $2a + 6 = 16$ **[1]**

32. Two identical packets balance with 14 identical balls. **[1]**

How many balls are equal to one packet?

33. A student writes: **[1]**

$2y = 12$

$y = 12 - 2$

Find the correct value of y.

34. Three identical boxes together weigh 21 kg. **[1]**

Find the weight of one box.

35. One bag and a 5 kg weight together weigh 17 kg. **[1]**

Find the weight of the bag.

Section B

36. A pen costs ₹ 15 more than a pencil. If 3 pens and 2 pencils cost ₹ 150, find the cost of each. **[2]**

37. On the left plate there are 2 identical sacks and 3 kg weight; on the right plate, there are 1 sack **[2]**

and a 5 kg weight. If the scale balances, find the weight of one sack. Then interpret this result as if the sacks contained grains - what does your answer mean?

38. Solve these equations: **[2]**

a. $2x + 1 = 6 - (2x - 3)$

b. $10 - 5x = 3(x - 4) - 2(x - 7)$

39. Solve the equation 2a + 5 = 17 and find the value of 4a. **[2]**

40. Three identical boxes and a 6 kg weight together weigh 24 kg; find the weight of one box and also find the total weight of five such boxes. **[2]**

41. A student solves the equation 4x = 20 and writes x = 20 - 4; write the correct value of x and state the correct operation that should be used. **[2]**

42. Solve the equation 5p - 3 = 22 and find the value of 2p - 1. **[2]**

43. A student solves 7y = 21 and writes y = 21 - 7; write the correct value of y and state the correct operation used to remove 7. **[2]**

44. Solve these equations: **[2]**

a. $5(r + 2) = 10$

b. $-3(u + 2) = 2(u - 1)$

45. Solve the equation 4(x - 1) = 12 and find the value of x - 2. **[2]**

46. Solve the following equation: 6 y + 7 = 4 y + 21. **[2]**

47. Solve these equations: **[2]**

a. $6(x - 1) = 2(x - 1) - 4$

b. $3 - 7s = 7 - 3s$

Section C

48. Seema is Anus younger sister. Seema is 4 years younger than Radha. Can you write Seema's age in terms of Anus age? Take Anus age to be x years. **[3]**

49. Solve 3m - 18 = 6 by trial and error method. **[3]**

50. State whether each is an equation or not: **[3]**

a. 3x + 5

b. 7 = 3x - 2

c. x + 4 > 10

51. A number trick says: Think of a number x. triple it and then add 12. After that half the result and subtract 3. You always get the same final expression. **[3]**

a. Write the final result in terms of x.

b. Show that the trick always gives an answer of (3/2)x + 3.

52. A father is 5 times as old as his son. After 18 years, he will be twice as old as his son. Find their present ages. **[3]**

53. Solve 5p + 2 = 17 by the trial and error method: **[3]**

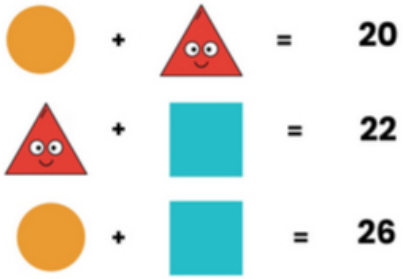
54. A rectangle has a perimeter of 96 cm. Its length is 8 cm more than twice its width. Find the dimensions of the rectangle. **[3]**

55. Write equation and solve **[3]**

a. Add 4 to one fourth of a number gives 20.

b. If you take away 5 from 5 times a number you get 50.

56. A puzzle board shows three equations using shapes: **[3]**



Find the value of Circle, Triangle, and Square.

57. Construct 3 equations starting with x = 5. **[3]**

Section D

58. **Read the following text carefully and answer the questions that follow:** **[4]**

A group of friends buys chocolates worth ₹180. Each friend pays the same amount. There are 4 more chocolates than the number of friends. Let the number of friends be x and the cost of one chocolate be ₹5.

Questions:

a. Write an expression for the number of chocolates bought. **(1)**

b. Write the total cost of chocolates using x. **(1)**

c. Form and solve an equation to find the number of friends. **(2)**

OR

How much does each friend pay? **(2)**

59. **Read the following text carefully and answer the questions that follow:** **[4]**

The perimeter of a rectangular garden is 60 m. Its length is 5 m more than its breadth. Let the breadth be x metres.

Questions:

a. Write an expression for the length of the garden. **(1)**

b. Write the formula for the perimeter of a rectangle. **(1)**

c. Form and solve an equation to find the breadth of the garden. **(2)**

OR

Find the length of the garden. **(2)**

60. **Read the following text carefully and answer the questions that follow:** **[4]**

A student saves the same amount every week. After 8 weeks, the total saving is ₹360. Let the weekly saving be ₹x.

Questions:

a. Write an equation representing the situation. **(1)**

b. What does the term 8x mean? **(1)**

c. Solve the equation to find the weekly saving. **(2)**

OR

How much money will the student save in 12 weeks at the same rate? **(2)**

Section E

61. Solve 3m - 14 = 4 by the trial and error method. **[5]**

62. The sum of three number is 15. If the second number is subtracted from the sum of first and third number then we get 5 an if twice of first number is added to second and if the third number is **[5]**

subtracted from the sum we get 4.

63. Write the following equations in statement forms: **[5]**

a. p + 4 = 15

b. m - 7 = 3

c. 2m = 7

d. m/5 = 3

e. (3m)/5 = 6

64. A father divides ₹5000 between his two children such that the elder child gets ₹x more than twice the younger child's amount. **[5]**

Also, 3 times the younger child's amount is 600 less than the elder child's amount. Find how much each child receives.

65. A rectangular garden is 8 m longer than its width. If its length is increased by 5 m and its width is increased by 3 m, the area becomes 455 m^2. Form an equation and find the original dimensions of the garden. **[5]**

66. A taxi service charges a fixed amount plus a rate per kilometre. A passenger travelled 12 km and paid ₹260. Another passenger travelled 18 km and paid ₹340. Let the fixed charge be F and the rate per km be R. Form equations and find the values of F and R. **[5]**

67. Write the equations for the following statements: **[5]**

a. The sum of the numbers x and 4 is 9.

b. 2 subtracted from y is 8.

c. Ten times a is 70.

d. The number b, when divided by 5, gives 6.

e. Three-fourths of t is 15.

68. A two-digit number is such that its tens digit is 3 more than its units digit. If the digits are reversed, the new number becomes 27 less than the original number. Find the number. **[5]**

69. Give the steps which are used to separate the variable and then solve the equation: **[5]**

a. 3n - 2 = 46

b. 5m + 7 = 17

Section F

70. **Fill in the blanks:** **[6]**

(a) The ancient Indian name for algebra is _____. **[1]**

(b) The solution of the equation **[1]**

x + 7 = 15

is x = _____.

(c) If $\frac{x}{2} = 9$, then $x =$ _____. **[1]**

(d) The value of m in **[1]**

4(m - 1) = 12

is _____.

(e) If 4 x = 36, then x = _____. **[1]**

(f) The word "algebra" comes from the Arabic word al-jabr used in the book of _____. **[1]**